

Relational σ -Additive Probability on Orthomodular Lattices: A Working Problem-Set

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Problem-set — June 2026

Abstract

A self-contained kit for attacking the descent-axis open problem under the *Reading-1* (relational = no hidden realisation) framing. Everything needed to sit down with pen and paper is defined inline: orthomodular lattices and the orthogonal/meet-zero gap; states, dispersion-free states, concreteness; contextuality and the Fine equivalence; the centre and the Stone/Loomis-Sikorski correspondence; and the construction $\coprod_n \text{MO}_2$. The open target is stated formally (§7), reduced to a single existence question, and — following the June 2026 analysis (§8) — further reduced to one deciding lemma about countable orthogonal families. The concrete sub-tasks (§9) carry the constraints established so far (the Wright bound; the off-centre requirement; the outcome-cardinality and tightness refinements). Companion narrative: `oml_onboarding.tex`, `reading1_prize_reduction.md`, `sigma_essential_nonemptiness_finding.md`, `sigma_essential_construction_attempt.md`. This document is presentation, not proof: open questions are flagged as open; included proofs are short and self-verifying or cited.

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Why this matters

The question behind the question. Classical probability is built on a sample space: a set Ω of definite states, with events as its subsets. A measure is then an average over those states. The programme this problem-set serves asks whether probability can be had *without* that scaffolding — *relationally*, from the entailment order of propositions alone, with no underlying space of

hidden definite states to descend from. In the Boolean (classical) case the answer is yes and well understood (this is the content of the CE / Paper-I results: coherence forces a unique σ -additive measure, no Ω assumed). The open frontier is the *non-Boolean* case, where some propositions are *incompatible* — not jointly decidable — so the event algebra is an orthomodular lattice rather than a Boolean one. There, classical probability has no room for the incompatibility, and whether an honest (σ -additive) relational probability theory exists at all is unsettled.

Why “contextual” is the precise form of “relational.” A measure that is secretly an average over hidden definite states is *not* relational — it is classical probability in disguise. By Fine’s theorem (Thm. 3.3) “average over hidden states” is exactly *non-contextual*. So the relational prize is a *contextual* σ -additive state: probability with no global hidden joint, genuinely on the non-distributive lattice. This is the measure-theoretic analogue of *localic* (point-free) topology — spaces recovered from their lattices of opens — but in the one regime localic measure theory does not reach: the *non-distributive* one.

What an answer decides. A *witness* (§9, Exit A) would be the first genuine relational σ -additive probability on a non-Boolean lattice — a new object, and a positive resolution of a two-year programme thread. An *impossibility proof* (Exit B) would be just as decisive in the other direction: it would show that the relational content the programme seeks is *operationally invisible* — present finitely, but unable to survive the passage to the countable level — turning a philosophical worry into a theorem. The finite case is already settled (Wright’s pentagon, §6); the entire live question is whether the phenomenon can be σ -essential, i.e. forced by the infinite structure and not visible in any finite piece. That “finite coherence vs. the countable commitment” shape is the same one at the heart of CE / Paper I — which is why this problem, though it wears quantum-logic clothing, rejoins the programme’s origin rather than departing from it.

1 Orthomodular lattices and the orthogonal/meet-zero gap

Definition 1.1 (Orthomodular lattice). An *orthomodular lattice* (OML) is a bounded lattice $(L, \wedge, \vee, 0, 1)$ with an order-reversing involution $a \mapsto a^\perp$ such that

$$a \vee a^\perp = 1, \quad a \wedge a^\perp = 0, \quad (a^\perp)^\perp = a,$$

and satisfying the *orthomodular law*:

$$a \leq b \implies b = a \vee (b \wedge a^\perp).$$

Write $a \perp b$ (“*a* orthogonal *b*”) for $a \leq b^\perp$. L is *Boolean* iff it is distributive.

Definition 1.2 (The two relations, and the gap). For $a, b \in L$:

$$a \perp b :\iff a \leq b^\perp \quad (\text{orthogonal}), \quad a \wedge b = 0 \quad (\text{meet-zero}).$$

Always $a \perp b \implies a \wedge b = 0$. The converse *fails* in a non-Boolean OML; the failure is the source of everything here.

Example 1.3 (MO_2 , the minimal gap). $\text{MO}_2 = \{0, 1, a, a^\perp, b, b^\perp\}$ (the “Chinese lantern”): four atoms, each pair of distinct atoms joining to 1 and meeting to 0. So $a \wedge b = 0$ but $a \not\leq b^\perp$ (as $b^\perp \in \{b^\perp\}$ and $a \neq b^\perp$), i.e. $a \wedge b = 0$ yet $a \not\perp b$. MO_2 is the smallest non-Boolean OML and the unit of every construction below.

Definition 1.4 (Completeness notions). L is σ -complete if every countable family $\{a_n\} \subseteq L$ has a join $\bigvee_n a_n \in L$. L is σ -orthocomplete if this holds for every countable *pairwise-orthogonal* family. (σ -complete $\implies \sigma$ -orthocomplete; the converse fails on non-Boolean OMLs.)

Definition 1.5 (Centre). $z \in L$ is *central* if it is compatible with every element and $L \cong [0, z] \times [0, z^\perp]$ as OMLs. The *centre* $Z(L) = \{\text{central elements}\}$ is a Boolean subalgebra; L is *irreducible* if $Z(L) = \{0, 1\}$.

2 States, dispersion-free states, concreteness

Definition 2.1 (State). A *state* on L is $s: L \rightarrow [0, 1]$ with $s(1) = 1$ and finite additivity on orthogonal pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$. It is σ -additive if $s(\bigvee_n a_n) = \sum_n s(a_n)$ for countable pairwise-orthogonal $\{a_n\}$ (join taken in L). A state is *dispersion-free* (2-valued) if $s(L) \subseteq \{0, 1\}$.

Definition 2.2 (Concrete logic). A set S of states is *order-determining* if $(\forall s \in S : s(a) \leq s(b)) \Rightarrow a \leq b$. L is *concrete* (equivalently *set-representable*) iff it admits an order-determining set of *dispersion-free* states (Gudder [4]). Equivalently, L embeds into some $(\mathcal{P}(X), \subseteq, X \setminus (-))$ preserving existing orthogonal joins.

Remark 2.3 (Separating vs. spanning — the load-bearing distinction). Concreteness asks the dispersion-free states to *separate* the order. It does *not* ask them to *span* the state space (i.e. that every state be a mixture of them). The whole problem lives in this gap; keep it in view.

Fact 2.4 (MO_2 has no dispersion-free homomorphism, but is concrete). MO_2 has exactly four order-determining dispersion-free states (each sending one atom-pair to 0 and the other to 1), so MO_2 is concrete. None is a lattice homomorphism: for the state s with $s(a) = s(b) = 0$ one has $s(a \vee b) = s(1) = 1 \neq 0 = \max(s(a), s(b))$ (equivalently $1 > 0 = s(a) + s(b)$, so s is not subadditive). Concrete uses states, not homomorphisms.

3 Contextuality and the Fine equivalence

The Reading-1 reading of “relational” — the measure carries *no hidden realisation* — is made precise by contextuality. Fix a concrete OML L with order-determining dispersion-free set S_{df} .

Definition 3.1 (Contextual state). A σ -additive state w on L is *non-contextual* if it lies in the closed convex hull $\overline{\text{conv}}(S_{\text{df}}^\sigma)$ of the σ -additive dispersion-free states $S_{\text{df}}^\sigma \subseteq S_{\text{df}}$ (in the topology of pointwise convergence on L); otherwise *contextual*.

Remark 3.2 (Why the hidden variables must be σ -additive — else the problem collapses). The restriction to S_{df}^σ (genuine σ -additive valuations, not merely finitely-additive “free-ultrafilter” points) is *forced* for the relational reading. S_{df} is *compact* (closed in $\{0, 1\}^L$ by Tychonoff; each $\hat{a} = \{s : s(a) = 1\}$ clopen), so a finitely-additive global section on the clopen field always upgrades to a σ -additive Radon measure on S_{df} by Carathéodory (continuity at $\emptyset$ is free on a compact space). Hence non-contextuality *over all of* S_{df} would hold automatically whenever it holds on every finite sub-OML — the problem would close trivially (Exit B by fiat), and the “hidden variables” would include the fake points the programme excludes (Rmk. 4.4). The whole open content is whether w can be carried by the *genuine* points S_{df}^σ .

Theorem 3.3 (Fine, in lattice form). *For a concrete L , the following are equivalent for a σ -additive state w :*

- (i) w is non-contextual ($w \in \overline{\text{conv}}(S_{\text{df}}^\sigma)$);
- (ii) $w = \int_{S_{\text{df}}^\sigma} s d\mu(s)$ for a probability measure μ concentrated on the σ -additive points S_{df}^σ (a global hidden joint by genuine points);
- (iii) the context-wise distributions w induces on the Boolean blocks of L admit a single global joint distribution reproducing all of them.

A dispersion-free state is a deterministic global section, so a global joint and a hull-decomposition are the same object (Fine [3]; Abramsky–Brandenburger [1]).

Remark 3.4 (Three point-spaces — do not conflate). The recurring trap. Distinguish:

- (A) the *external* points (e.g. Hilbert rays of H for $L(H)$): an auxiliary space the measure could be reduced to (Gleason’s ρ in $s = \text{tr}(\rho \cdot)$). Reading-1 *forbids* reduction to these.

- (B) the *canonical dual* points $P(A) = \{\uparrow p : p \in L\}$ (principal filters — one per *element*): built from L , hence *permitted*.
- (C) the *dispersion-free states* S_{df} (one per consistent global *valuation*): the hidden-variable points of Def. 3.1.

“Relational = contextual” is a condition on (C), *not* on (B): a witness is *point-rich* on $P(A)$ yet has no global S_{df} -joint. (A), (B), (C) are genuinely different objects.

The open target. Relational σ -additive probability = a contextual σ -additive state on a concrete OML. (The formal open problem is §7.)

4 The Boolean engine, and why it does not transfer

Theorem 4.1 (Stone / Loomis–Sikorski). *For a Boolean algebra B with Stone space $\text{St}(B)$: B is σ -complete iff $\text{St}(B)$ is basically disconnected (the closure of every cozero set is open); and a finitely-additive state on B is σ -additive iff, viewed as a measure on $\text{St}(B)$, it concentrates on the principal (isolated/atomic) part. In particular a σ -additive state on $\mathcal{P}(\mathbb{N})$ is atomic: $w = \sum_n w(\{n\})\delta_n$.*

Theorem 4.2 (Pták–Pulmannová [7]). *An OML is Boolean iff it carries a unital set of subadditive states. Hence the Boolean-forcing property is subadditivity, not σ -additivity: σ -additivity alone does not collapse a non-Boolean OML, so non-Boolean σ -additive states genuinely exist.*

Remark 4.3 (The Boolean kill-mechanism, and its Boolean-only hinge). On a Boolean algebra the chain “ σ -additive $\Rightarrow$ concentrates on principal ultrafilters $\Rightarrow$ non-contextual” is forced, because there

$$\text{principal ultrafilter} = \text{atom} = \text{dispersion-free state}.$$

This triple identity is *Boolean-only*. On an OML with centre $Z(L)$: w restricted to $Z(L)$ is a σ -additive Boolean measure, so (by Thm. 4.1) it concentrates on the *central atoms* — but a central atom z subtends a (possibly non-Boolean) block $[0, z]$, and a state with $w(z) = 1$ can be *contextual within* $[0, z]$. So

$$\underbrace{\sigma\text{-add} \Rightarrow \text{concentrate on central atoms}}_{\text{transfers}}, \quad \underbrace{\text{concentrate} \Rightarrow \text{non-contextual}}_{\text{Boolean-only; FAILS}}.$$

For *irreducible* L ($Z(L) = \{0, 1\}$) the centre argument says only $w(1) = 1$ — vacuous. *The Boolean kill-mechanism evaporates exactly off-centre, where a witness would live.* (Routing verdict: `sigma_essential_nonemptiness_finding.md`.)

Remark 4.4 (Why no faithful set representation — the universal floor). A faithful σ -tribe / Loomis–Sikorski representation realises L as a field of sets, preserving $\wedge, \vee$ as $\cap, \cup$; its image is a sublattice of a Boolean algebra, hence distributive, forcing L Boolean. So *no* non-Boolean OML admits one. A relational measure can therefore never arise from a faithful descent to points — the target is necessarily “point-free” in the sense of Rmk. 3.4(A), *not* of (B).

5 The construction $\prod_n \text{MO}_2$, and what it settles

Definition 5.1 ($\prod_n \text{MO}_2$). The full direct product of countably many copies of MO_2 , coordinatewise operations. It is concrete (coordinate states separate), σ -complete (product of finite complete lattices), non-Boolean, infinite. Its centre contains $\prod_n \{0, 1\} \cong \mathcal{P}(\mathbb{N})$.

Proposition 5.2 (The interleaving property (★) is cheap). *There exist p and an infinite orthogonal family $\{a_n\}$ in $\prod_n \text{MO}_2$ with $p \wedge a_n = 0$ and $p \not\leq a_n$ for all n . (Take $a_n = a$ in coordinate n , 0 elsewhere; $b_n = b$ in coordinate n , 0 elsewhere; $p = \bigvee_n b_n = (b, b, b, \dots)$, a genuine orthogonal join. Coordinate n : $a \wedge b = 0$ but $b \not\leq a^\perp$.) So “(★) + concrete + σ -complete + non-Boolean” is satisfied by a one-line object: it is not the discriminator.*

Proposition 5.3 ($\prod_n \text{MO}_2$ is not a witness — contextuality is only finitely additive). *Every σ -additive state on $\prod_n \text{MO}_2$ is non-contextual (in $\overline{\text{conv}}(S_{\text{df}})$). Sketch: a contextual state would have to act diffusely on the central $\mathcal{P}(\mathbb{N})$ (a free-ultrafilter/Banach-limit type state); but by Thm. 4.1 a σ -additive state on $\mathcal{P}(\mathbb{N})$ is atomic, hence concentrates on coordinates, hence is a countable mixture of coordinate dispersion-free states. The contextual states exist only in the finitely additive layer; σ -additivity kills them. (This is the CE / Paper-I compactness phenomenon, and the reason $\prod_n \text{MO}_2$ fails: all its infinitary structure is central.)*

6 The bound: Wright’s pentagon

Theorem 6.1 (Wright [8]). *There is a finite, concrete OML — the pentagon (pentagram) logic, a Greechie loop of length five (hence a lattice) — with an order-determining family of dispersion-free states, carrying a state not in their convex hull, i.e. a contextual state.*

Remark 6.2 (What Wright settles, and what it leaves). A finite OML is vacuously σ -complete and its states vacuously σ -additive. So “concrete σ -complete non-Boolean OML with a σ -additive contextual state” is *already inhabited* by Wright — and the general spanning theorem “every state is a dispersion-free mixture” is *false*. What Wright does *not* give: contextuality requiring the *countable* structure. Its contextuality is witnessed by a finite sub-OML (itself). That single missing qualifier — **σ -essential** — is the entire open content.

7 The open target

*The open target. (**σ -essential contextuality**).* Does there exist a concrete, σ -complete, non-Boolean OML L and a σ -additive state w on L such that:

- (1) w is *contextual* — $w \notin \overline{\text{conv}}(S_{\text{df}})$ (no global hidden joint over the dispersion-free states); and
- (2) the contextuality is *σ -essential* — the restriction of w to *every finite sub-OML* of L is non-contextual (so the obstruction lives only in the countable structure)?

Equivalently: a relational σ -additive probability whose relationality is not already present finitely. **Or:** prove no such (L, w) exists.

Remark 7.1 (Status). **Well-posed, principled, uninhabited.** No witness ($\prod_n \text{MO}_2$ fails by Prop. 5.3; Wright is finite, Rmk. 6.2); no impossibility proof. The CE-routing analysis (Rmk. 4.3) returned *GAP*: CE neither forbids nor produces a witness. The June 2026 analysis (§8) localises both exits to a *single* lemma.

8 Reduction: from gluing to section σ -tightness (June 2026)

This section records the sharpening of the open target into a single decidable lemma. It supersedes the earlier “paste Wright blocks” programme: that route is *dead* (Rmk. 8.1). Everything here is verified by hand or cited.

Remark 8.1 (Loop-pasting is dead). Contextuality of a Greechie loop L_n (n Boolean blocks pasted cyclically) is a *parity* phenomenon, not a length threshold: the dispersion-free states fail to span iff n is *odd*, for *every* odd n (hull bound = independence number $\alpha(C_n) = \lfloor n/2 \rfloor$; Wright’s $\frac{1}{2}$ -on-intertwines state needs $n/2$; $n/2 > (n-1)/2$ iff n odd). So a tower of odd loops is contextual at *every finite stage* (Reading-1 fails; same species as $\prod_n \text{MO}_2$ ’s finitely-additive contextuality), and even loops never span-fail. The loop-parity effect is a *finite, finitely-additive* witness — never σ -essential. Pasting contextual blocks cannot manufacture contextuality-only-at-infinity.

Proposition 8.2 (The finitely-additive global section always exists). *Let w be a σ -additive state on a concrete OML L , non-contextual on every finite sub-OML. Then a finitely-additive global hidden joint μ exists (a charge on the event algebra A marginalising to w on every context).*
 Proof. *The finitely-additive probability charges on A form a weak-* compact set (Banach–Alaoglu in $[0, 1]^A$). For each finite sub-OML F , the set $K_F = \{\mu : \mu|_F = w|_F\}$ is weak-* closed; finite-local-non-contextuality makes each K_F nonempty and any finite intersection (a single larger finite sub-OML) nonempty. Compactness + finite-intersection-property gives $\bigcap_F K_F \neq \emptyset$. $\square$ Note the compactness is of the **charge** space $\text{ba}(A)$, not of the assignment space $\Lambda = \prod_i O_i$; the former holds regardless of outcome cardinality. So the obstruction sits cleanly and solely at the σ -additivity upgrade of μ .*

Remark 8.3 (The intra-context horn is barred). A σ -additive w restricts to a σ -additive state on every sub- σ -algebra, hence $w|_C$ is σ -additive on every context C . The global μ 's marginal on C is $w|_C$. So μ 's per-context marginals are automatically σ -additive: *the σ -failure of μ can never be visible in a single context's marginal*. This kills the “decorative- σ ” candidate (a density/free-ultrafilter charge dressed as a scenario): it would force the σ -failure into one Boolean marginal, breaking the σ -additivity of w . The trilemma (central | intra-context | cross-contextual) collapses to a dilemma: *neither central (Rmk. 4.3) nor intra-context*.

Remark 8.4 (The gap is a non-tight projective family). By Prop. 8.2 and Rmk. 8.3, a witness is exactly: μ a finitely-additive joint with σ -additive context-marginals but *no* σ -additive version — a *non-tight projective family* (Daniell–Kolmogorov extension failing for want of tightness, the regime where the assignment space Λ is non-compact: countably/uncountably many outcomes per measurement). The one rigorous treatment of infinite AB scenarios (Barbosa–Abramsky, continuous-variable [2]) *defines* global sections as *Radon* (tight $\Rightarrow$ σ -additive) measures — assuming the gap shut by fiat. *Verify* this Radon restriction directly in their §3–4 before leaning on it. The gate: classical non-tight families are Boolean (factor through a product = one distributive context); the witness needs non-tightness *coupled* to non-distributivity, not merely *adjacent* to it.

Lemma 8.5 (A single orthogonal family is always block-contained — hence *not* the locus). *Every countable orthogonal family $\{a_i\}$ in a concrete σ -complete OML is contained in a Boolean sub- σ -algebra (a block).* Proof. *Orthogonality survives countable joins: $a \leq a_i^\perp \ \forall i \Rightarrow a \leq \bigwedge_i a_i^\perp = (\bigvee_i a_i)^\perp$. So $S \mapsto \bigvee_{i \in S} a_i$ ($S \subseteq \mathbb{N}$) is a Boolean σ -homomorphism into L with mutually orthogonal generators; the elements are pairwise compatible (Foulis–Holland), so they generate a Boolean sub- σ -algebra, which extends to a block by Zorn.* $\square$

Remark 8.6 (Why this is the wrong locus — correcting an earlier framing). An earlier draft posed “can an orthogonal family escape every block?” as the deciding question. Lem. 8.5 answers it *yes, near-trivially* — and that resolves *nothing*. A state restricted to one orthogonal family is a classical distribution over orthogonal outcomes, hence *never* contextual: the σ -failure cannot live inside a single family. The σ -condition is *constrained on* orthogonal families (Def. 2.1); the contextuality *lives on* the incompatible *overlap between distinct* families. The locus is the **gluing**, not any one family.

Remark 8.7 (The locus is the gluing across contexts). So the obstruction, if any, is in the σ -additive *global* upgrade across the cover of contexts — the finitely-additive joint always exists (Prop. 8.2) and the per-context marginals are forced σ -additive (Rmk. 8.3). The precise placement of that upgrade question is Target 8 below; first a lemma that clears away a *wrong* placement (F1).

Lemma 8.8 (Compatibility survives countable joins — F1 holds in *every* σ -complete OML). *Let L be σ -complete, $b \in L$, and $\{a_i\}$ a countable pairwise-compatible family with $b \leq a_i$ for all i . Then $b \leq \bigvee_i a_i$.* Proof. *Put $u = \bigvee_i (a_i \wedge b) \leq b$ and $v = \bigvee_i (a_i \wedge b^\perp) \leq b^\perp$ (exist by*

σ -completeness). Then $\bigvee_i a_i = u \vee v$: “ $\leq$ ” since $a_i = (a_i \wedge b) \vee (a_i \wedge b^\perp)$ (as $b C a_i$); “ $\geq$ ” since $a_i \wedge b$, $a_i \wedge b^\perp \leq \bigvee_i a_i$. Now b commutes with u ($u \leq b$) and with v ($v \leq b^\perp$, so $v \perp b$); by Foulis–Holland $b \wedge (u \vee v) = (b \wedge u) \vee (b \wedge v) = u \vee 0 = u$ and dually $b^\perp \wedge (u \vee v) = v$, whence $(p \wedge b) \vee (p \wedge b^\perp) = u \vee v = p$ for $p = \bigvee_i a_i$. $\square$

Remark 8.9 (F1 is true but does *not* discriminate — correcting a detour). An earlier draft made F1 (“blocks are maximal Boolean σ -subalgebras”) the pivot. Lem. 8.8 settles it *true in all σ -complete OMLs* (the commutant $\{a : a C b\}$ is a σ -sub-OML; Kalmbach [5]) — *using no concreteness and no σ -additive states*. And F1 *does not imply Exit B*. The conflation: gluing σ -additive block-marginals into a global *state on L* is trivial — that state is just w . The contextuality question is gluing into a global *section on S_{df}^σ* (a σ -additive *mixture of σ -additive dispersion-free states*, i.e. *spanning by genuine points*, Thm. 3.3). F1 is a *within-block* fact and bears only on the former. *Independence, sharply*: on $L(H)$ every block is σ -closed (commuting projections generate a von Neumann algebra, F1 holds) yet there are *no* dispersion-free states at all (Kochen–Specker) — F1 holds, spanning fails maximally. So $\text{F1} \not\Rightarrow \text{Exit B}$. Drop F1 as the gate.

Proposition 8.10 (The σ -representing measure on S_{df} always exists — the upgrade is *free*). S_{df} is compact (closed in $\{0,1\}^L$ by Tychonoff — each orthogonal-pair additivity condition constrains three coordinates, hence is closed; each $\hat{a} = \{s : s(a) = 1\}$ is clopen). Hence the finitely-additive section μ of Prop. 8.2, defined on the clopen field $\langle \hat{a} \rangle$, has continuity at $\emptyset$ for free (clopen $F_n \downarrow \emptyset$ are eventually empty by compactness) and extends, by Carathéodory, to a σ -additive Radon measure on S_{df} with $\mu(\hat{a}) = w(a)$. So a σ -additive global section on S_{df} always exists — whenever w is non-contextual on every finite sub-OML. The “ σ -upgrade” is not the obstruction.

The open target (The real discriminator: support on the *genuine points* S_{df}^σ). By Prop. 8.10, w is always a σ -additive mixture of *some* dispersion-free states — but μ lives on *all* of S_{df} , including the merely finitely-additive “fake points” $S_{\text{df}} \setminus S_{\text{df}}^\sigma$. For each countable orthogonal family with join p , the gap set $G = \hat{p} \setminus \bigcup_i \hat{a}_i = \{s : s(p) = 1, s(a_i) = 0 \forall i\}$ is closed and μ -null ($\mu(\hat{p}) = w(p) = \sum w(a_i) = \mu(\bigcup_i \hat{a}_i)$). But $S_{\text{df}} \setminus S_{\text{df}}^\sigma = \bigcup_{\text{families}} G$ is an **uncountable** union of μ -null sets, which may carry positive measure; and S_{df}^σ (an intersection of uncountably many opens) is generally non-Borel.

- **Exit A (witness)** $\Leftrightarrow$ a σ -additive w whose *every* representing μ must charge $S_{\text{df}} \setminus S_{\text{df}}^\sigma$ (w not carriable by genuine σ -additive hidden variables).

- **Exit B (impossibility)** $\Leftrightarrow$ every such w admits a μ with inner measure $\mu_*(S_{\text{df}}^\sigma) = 1$. This is a *support-on-a-non-Borel-conull-set* / additivity-of-the-null-ideal (add(null), real-valued-measurable-cardinal) question. Infinite outcomes are what make it non-vacuous: an *infinite block* makes σ -additivity a real constraint on dispersion-free states, i.e. makes $S_{\text{df}}^\sigma \subsetneq S_{\text{df}}$ possible. $L(H)$ is not concrete, so the concrete case is genuinely open.

9 Sub-tasks — the pen-and-paper frontier

The deciding question (Target 8) is the frontier; the sub-tasks below are its two horns. Standing checkpoints throughout: the witness OML must remain σ -complete and *irreducible*, and must be *infinite-outcome* (non-compact Λ , Rmk. 8.4); do not drop these in the scenario-language translation. The constraints of Rmk. 4.3, Prop. 5.3, and §8 prune the search.

Exit A — build a σ -essential witness

Sub-task 1 (A w supported off the genuine points). Exhibit a concrete σ -complete *irreducible* OML with an *infinite block* and a σ -additive state w , non-contextual on every finite sub-OML,

such that *every* representing measure μ on S_{df} (one always exists, Prop. 8.10) must charge the fake points $S_{\text{df}} \setminus S_{\text{df}}^\sigma$ — i.e. w is *not* a σ -additive mixture of σ -additive dispersion-free states (Target 8, Exit A). *Why this suffices:* that is exactly a σ -additive relational state with no genuine hidden-variable realisation. *Where to look:* the infinite block makes $S_{\text{df}}^\sigma \subsetneq S_{\text{df}}$ (Rmk. 3.2); the task is to force μ 's mass onto the non-Borel conull complement — an add(null)/measure-concentration construction — *not* pasted contextual blocks (Rmk. 8.1), *not* products of finite blocks (off-centre, Rmk. 4.3), *not* a single orthogonal family (Rmk. 8.6), *not* an intra-block density charge (Rmk. 8.3).

Sub-task 2 (Off-centre, infinite-outcome). Maintain that the infinite non-Boolean structure sits *off-centre* (L irreducible, or relevant infinite blocks non-central): if all infinitary structure is central, Stone over the centre (Thm. 4.1) forces atomicity and kills contextuality σ -additively (Prop. 5.3). Combined with Rmk. 8.4, the infinite-dimensionality must enter through *outcome multiplicity*, not mere lattice size. *Excludes products of finite blocks and finite-outcome colimits a priori.*

Exit B — prove no witness exists

Sub-task 3 (A genuine-point support theorem for concrete OMLs). Prove the “support” horn of Target 8: every σ -additive state on a concrete σ -complete OML admits a representing measure μ on S_{df} with inner measure $\mu_*(S_{\text{df}}^\sigma) = 1$ — w is carried by the *genuine* (σ -additive) dispersion-free points. *Why this suffices:* σ -essential contextuality is then impossible. **Constraint:** this is *not* F1 (Lem. 8.8, the within-block fact, is already true and does not suffice, Rmk. 8.9), and *not* the σ -upgrade on S_{df} (free, Prop. 8.10); the content is that μ can be *concentrated* on the non-Borel conull set S_{df}^σ — a measure-support / add(null) statement where the fake points carry no forced mass. Nor may it reuse the $\prod_n \text{MO}_2$ /Stone-over-centre mechanism (vacuous off-centre, Rmk. 4.3). *Such a theorem reinstates the empiricist-underdetermination conclusion as a theorem* (direction2_gate_finding.md).

Remark 9.1 (Discipline). Neither exit is decidable by a quick computation; both are Phase-4 mathematics. The recurring failure mode in this material is the clean “ X , therefore Y ” step that needs a qualifier (the Boolean triple identity of Rmk. 4.3 is the paradigm). Verify transfers; do not assume them. A candidate witness is checked by: is its claimed contextual state genuinely outside $\overline{\text{conv}}(S_{\text{df}})$, and is the obstruction genuinely absent from every finite sub-OML?

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